

Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

| Algorithm  | Ranking |
|------------|---------|
| Random     | 9.375   |
| NN         | 6.5625  |
| Sweep      | 8.5625  |
| SaveSeq    | 2.625   |
| SaveParall | 2.75    |
| MJ         | 2.375   |
| CMT        | 2.75    |
| Kilby      | 6.625   |
| SaveMatch  | 6.5     |
| FJ         | 6.875   |

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 9 degrees of freedom): 54.838636.  
P-value computed by Friedman Test: 0.

## 2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

| $i$ | algorithm  | $z = (R_0 - R_i)/SE$ | $p$      | Holm     | Finner   | Li       |
|-----|------------|----------------------|----------|----------|----------|----------|
| 9   | Random     | 4.624048             | 0.000004 | 0.005556 | 0.005683 | 0.006904 |
| 8   | Sweep      | 4.087328             | 0.000044 | 0.00625  | 0.011334 | 0.006904 |
| 7   | FJ         | 2.972602             | 0.002953 | 0.007143 | 0.016952 | 0.006904 |
| 6   | Kilby      | 2.807458             | 0.004993 | 0.008333 | 0.022539 | 0.006904 |
| 5   | NN         | 2.766171             | 0.005672 | 0.01     | 0.028094 | 0.006904 |
| 4   | SaveMatch  | 2.724885             | 0.006432 | 0.0125   | 0.033617 | 0.006904 |
| 3   | SaveParall | 0.247717             | 0.804353 | 0.016667 | 0.039109 | 0.006904 |
| 2   | CMT        | 0.247717             | 0.804353 | 0.025    | 0.04457  | 0.006904 |
| 1   | SaveSeq    | 0.165145             | 0.86883  | 0.05     | 0.05     | 0.05     |

Table 2: Post Hoc comparison Table for  $\alpha = 0.05$  (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value  $\leq 0.016667$ .

Finner's procedure rejects those hypotheses that have an unadjusted p-value  $\leq 0.039109$ .

Li's procedure rejects those hypotheses that have an unadjusted p-value  $\leq 0.006904$ .

### 3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

| i | algorithm  | unadjusted $p$ | $p_{Holm}$ |
|---|------------|----------------|------------|
| 1 | Random     | 0.00004        | 0.000034   |
| 2 | Sweep      | 0.000044       | 0.000349   |
| 3 | FJ         | 0.002953       | 0.02067    |
| 4 | Kilby      | 0.004993       | 0.029961   |
| 5 | NN         | 0.005672       | 0.029961   |
| 6 | SaveMatch  | 0.006432       | 0.029961   |
| 7 | SaveParall | 0.804353       | 2.41306    |
| 8 | CMT        | 0.804353       | 2.41306    |
| 9 | SaveSeq    | 0.86883        | 2.41306    |

Table 3: Adjusted  $p$ -values (FRIEDMAN) (I)

| i | algorithm  | unadjusted $p$ | $p_{Finner}$ | $p_{Li}$ |
|---|------------|----------------|--------------|----------|
| 1 | Random     | 0.00004        | 0.000034     | 0.000029 |
| 2 | Sweep      | 0.000044       | 0.000196     | 0.000333 |
| 3 | FJ         | 0.002953       | 0.008832     | 0.022016 |
| 4 | Kilby      | 0.004993       | 0.0112       | 0.036672 |
| 5 | NN         | 0.005672       | 0.0112       | 0.041448 |
| 6 | SaveMatch  | 0.006432       | 0.0112       | 0.046746 |
| 7 | SaveParall | 0.804353       | 0.877246     | 0.85979  |
| 8 | CMT        | 0.804353       | 0.877246     | 0.85979  |
| 9 | SaveSeq    | 0.86883        | 0.877246     | 0.86883  |

Table 4: Adjusted  $p$ -values (FRIEDMAN) (II)